

### 13. Dataset: Geographic Data

**City Population Avg. Temperature Elevation**

**City A 500000 75 1000**

**City B 700000 68 800**

**City C 600000 80 1200**

**1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.**

```
library(ggplot2)
```

```
library(maps)
```

```
geo$Longitude <- c(-120, -100, -80)
```

```
geo$Latitude <- c(35, 40, 45)
```

```
world <- map_data("world")
```

```
ggplot() +
```

```
  geom_map(
```

```
    data = world,
```

```
    map = world,
```

```
    aes(long, lat, map_id = region),
```

```
    fill = "lightgray",
```

```
    color = "white"
```

```
  ) +
```

```
  geom_point(
```

```
    data = geo,
```

```
    aes(x = Longitude, y = Latitude, size = Population),
```

```
    color = "red"
```

```
  ) +
```

```
  geom_text(
```

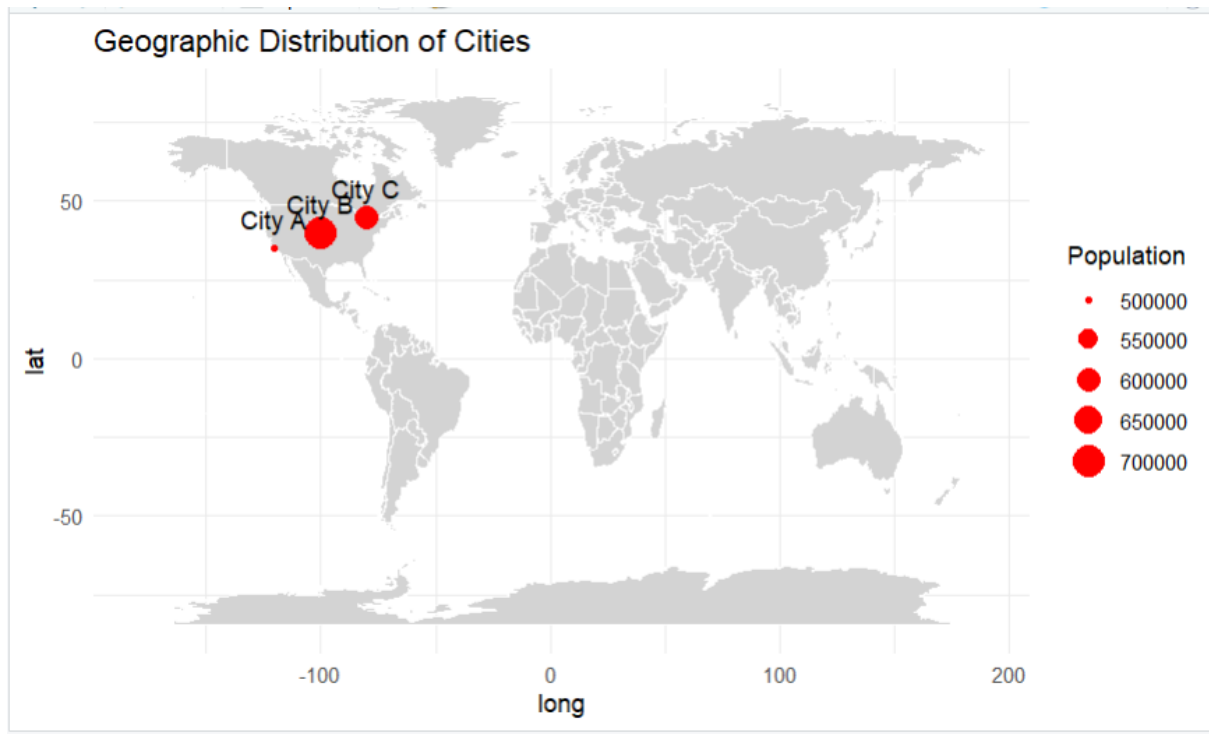
```
    data = geo,
```

```
    aes(x = Longitude, y = Latitude, label = City),
```

```

    vjust = -1
  ) +
  labs(
    title = "Geographic Distribution of Cities"
  ) +
  theme_minimal()

```



## 2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```

library(ggplot2)

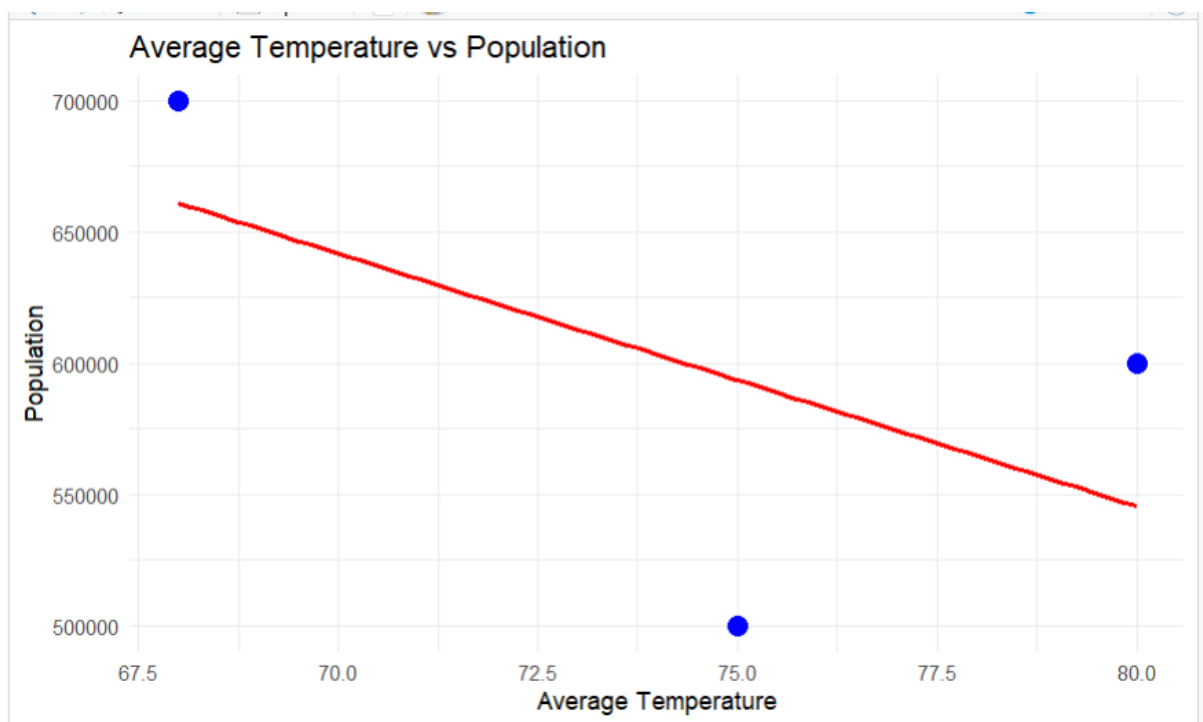
ggplot(geographic_data,
  aes(x = Avg_Temperature,
    y = Population)) +
  geom_point(
    size = 4,
    color = "blue"
  ) +
  geom_smooth(
    method = "lm",

```

```

se = FALSE,
color = "red"
) +
labs(
  title = "Average Temperature vs Population",
  x = "Average Temperature",
  y = "Population"
) +
theme_minimal()

```



### 3. Build a table to display the geographic data for each city. Label the table elements.

# Display the geographic data table

View(geographic\_data)

|   | City   | Population | Avg_Temperature | Elevation | Longitude | Latitude |
|---|--------|------------|-----------------|-----------|-----------|----------|
| 1 | City A | 5e+05      | 75              | 1000      | -120      | 35       |
| 2 | City B | 7e+05      | 68              | 800       | -100      | 40       |
| 3 | City C | 6e+05      | 80              | 1200      | -80       | 45       |

## 14. Dataset: Survey Results

Respondent Question 1 Question 2 Question 3

**1 A B C**

**2 B A D**

**3 C A B**

**1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.**

```
library(ggplot2)
```

```
library(reshape2)
```

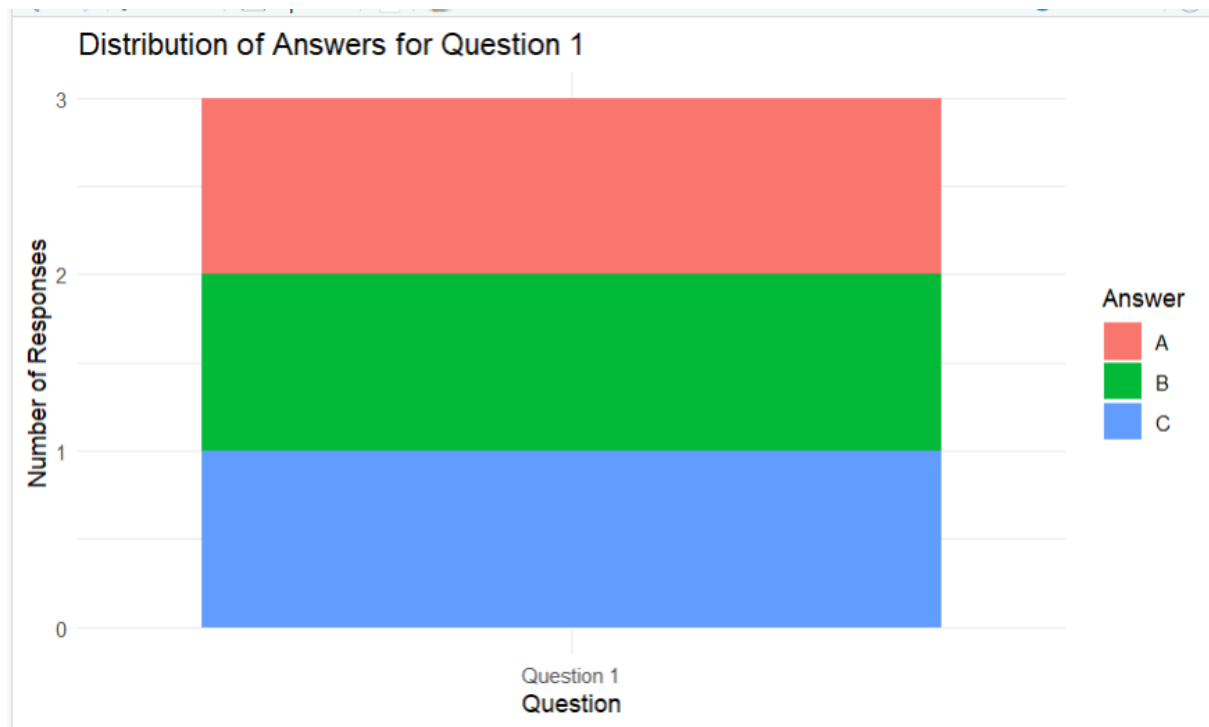
```
# Convert data to long format
```

```
long <- melt(survey,  
             id.vars = "Respondent",  
             variable.name = "Question",  
             value.name = "Answer")
```

```
# Stacked Bar Chart for Question 1
```

```
q1 <- subset(long, Question == "Question1")
```

```
ggplot(q1,  
       aes(x = "Question 1",  
           fill = Answer)) +  
geom_bar() +  
labs(  
  title = "Distribution of Answers for Question 1",  
  x = "Question",  
  y = "Number of Responses"  
) +  
theme_minimal()
```



## 2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

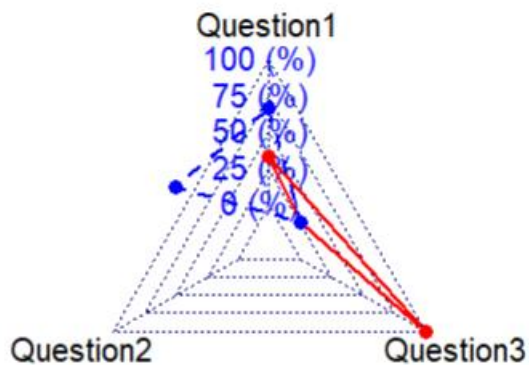
```
score <- data.frame(
  Respondent = survey$Respondent,
  Question1 = c(1,2,3),
  Question2 = c(2,1,1),
  Question3 = c(3,4,2)
)
```

```
radar <- data.frame(
  Question1 = c(4,1,2,3),
  Question2 = c(4,2,1,1),
  Question3 = c(4,3,4,2)
)
```

```
rownames(radar) <- c("Max", "Respondent1", "Respondent2", "Respondent3")
```

```
fmsb::radarchart(
  radar,
```

```
axistype = 1,
pcol = c("red","blue","green"),
plwd = 2
)
```



**3. Build a table to show the survey response data for each respondent. Label the table elements.**

```
View(survey)
```

# or

```
print(survey)
```

|   | Respondent | Question1 | Question2 | Question3 |
|---|------------|-----------|-----------|-----------|
| 1 | 1          | A         | B         | C         |
| 2 | 2          | B         | A         | D         |
| 3 | 3          | C         | A         | B         |

## Set 15: Product Sales Analysis

### Dataset: Monthly Product Sales

**Product ID Product Name January Sales February Sales March Sales**

**1 Product A 2000 2200 2400**

**2 Product B 1500 1800 1600**

**3 Product C 1200 1400 1100**

**1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.**

```
long <- melt(
  product_sales,
```

```
id.vars = c("Product_ID", "Product_Name"),
variable.name = "Month",
value.name = "Sales"
)
```

```
ggplot(long,
  aes(x = Product_Name,
      y = Sales,
      fill = Month)) +
geom_bar(stat = "identity",
  position = "dodge") +
labs(
  title = "Quarter 1 Product Sales",
  x = "Product Name",
  y = "Sales"
) +
theme_minimal()
```

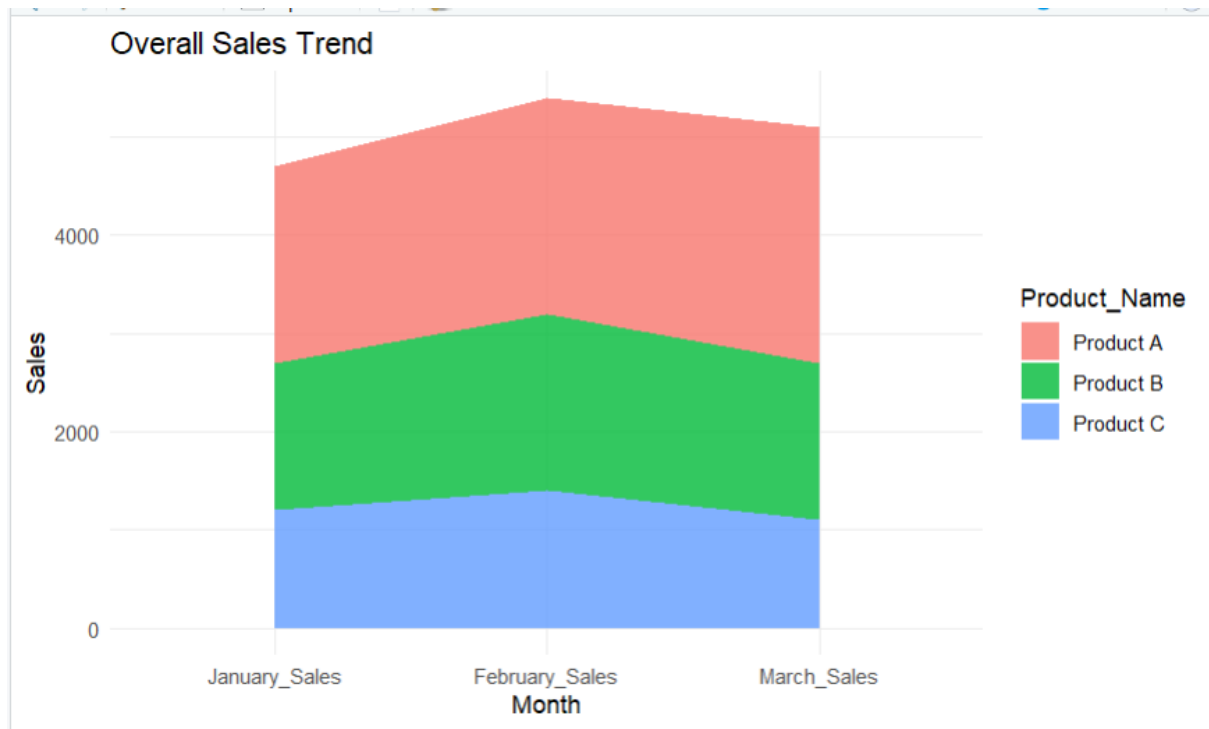


**2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.**

```

ggplot(long,
  aes(x = Month,
      y = Sales,
      fill = Product_Name,
      group = Product_Name)) +
  geom_area(alpha = 0.8) +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()

```



**3. Build a table to show the monthly sales data for each product. Label the table elements.**

```
View(product_sales)
```

# OR

```
print(product_sales)
```

|   | Product_ID | Product_Name | January_Sales | February_Sales | March_Sales |  | 5 columns                                                            |
|---|------------|--------------|---------------|----------------|-------------|--|----------------------------------------------------------------------|
| 1 | 1          | Product A    | 2000          | 2200           | 2400        |  | <div>Product_ID</div> <div>[1, 3]</div> <div>0% NA</div>             |
| 2 | 2          | Product B    | 1500          | 1800           | 1600        |  | <div>Product_Name</div> <div>3 unique</div> <div>0% NA</div>         |
| 3 | 3          | Product C    | 1200          | 1400           | 1100        |  | <div>January_Sales</div> <div>[1,200, 2,000]</div> <div>0% NA</div>  |
|   |            |              |               |                |             |  | <div>February_Sales</div> <div>[1,400, 2,200]</div> <div>0% NA</div> |
|   |            |              |               |                |             |  | <div>March_Sales</div> <div>[1,100, 2,400]</div> <div>0% NA</div>    |

Showing 1 to 3 of 3 entries, 5 total columns

## Set 16: Customer Demographics Analysis

### Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

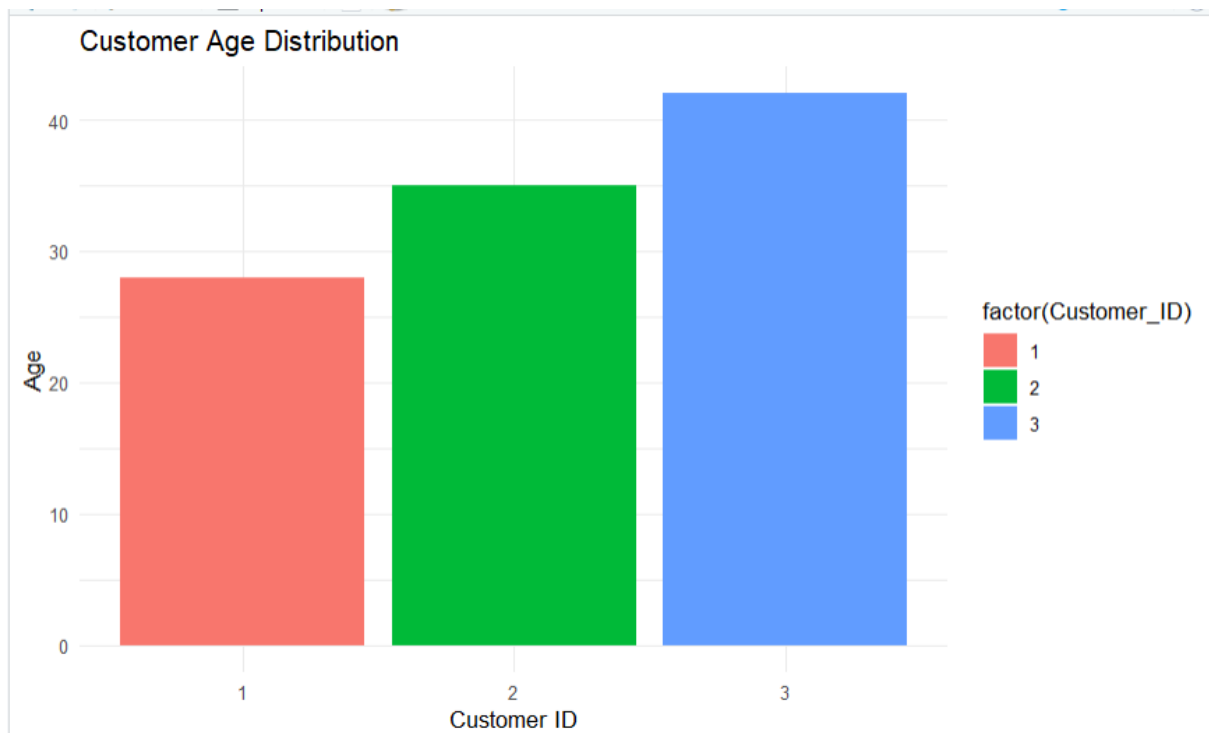
1 28 Female 50000

2 35 Male 60000

3 42 Female 75000

**1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.**

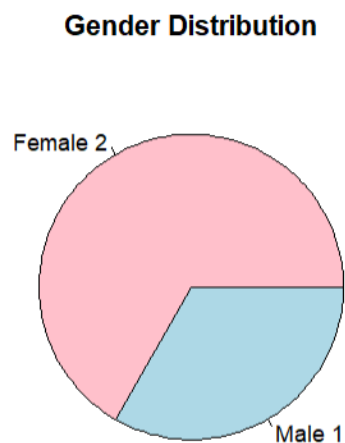
```
ggplot(customer,
  aes(x = factor(Customer_ID),
    y = Age,
    fill = factor(Customer_ID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
```



## 2. Generate a pie chart to represent the distribution of customers by gender.

```
gender_count <- table(customer$Gender)
```

```
pie(
  gender_count,
  labels = paste(names(gender_count), gender_count),
  col = c("pink", "lightblue"),
  main = "Gender Distribution"
)
```



### 3. Build a table to show the demographic information for each customer. Label the table elements.

```
View(customer)
```

```
# OR
```

```
print(customer)
```

|   | Customer_ID | Age | Gender | Income |
|---|-------------|-----|--------|--------|
| 1 | 1           | 28  | Female | 50000  |
| 2 | 2           | 35  | Male   | 60000  |
| 3 | 3           | 42  | Female | 75000  |

4 columns

Customer\_ID<dbl>

[1, 3]0% NA

Age<dbl>

[28, 42]0% NA

Gender<chr>

2 unique0% NA

Income<dbl>

[50,000, 75,000]0% NA

## Set 17: Employee Performance Analysis

### Dataset: Employee Performance

#### Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

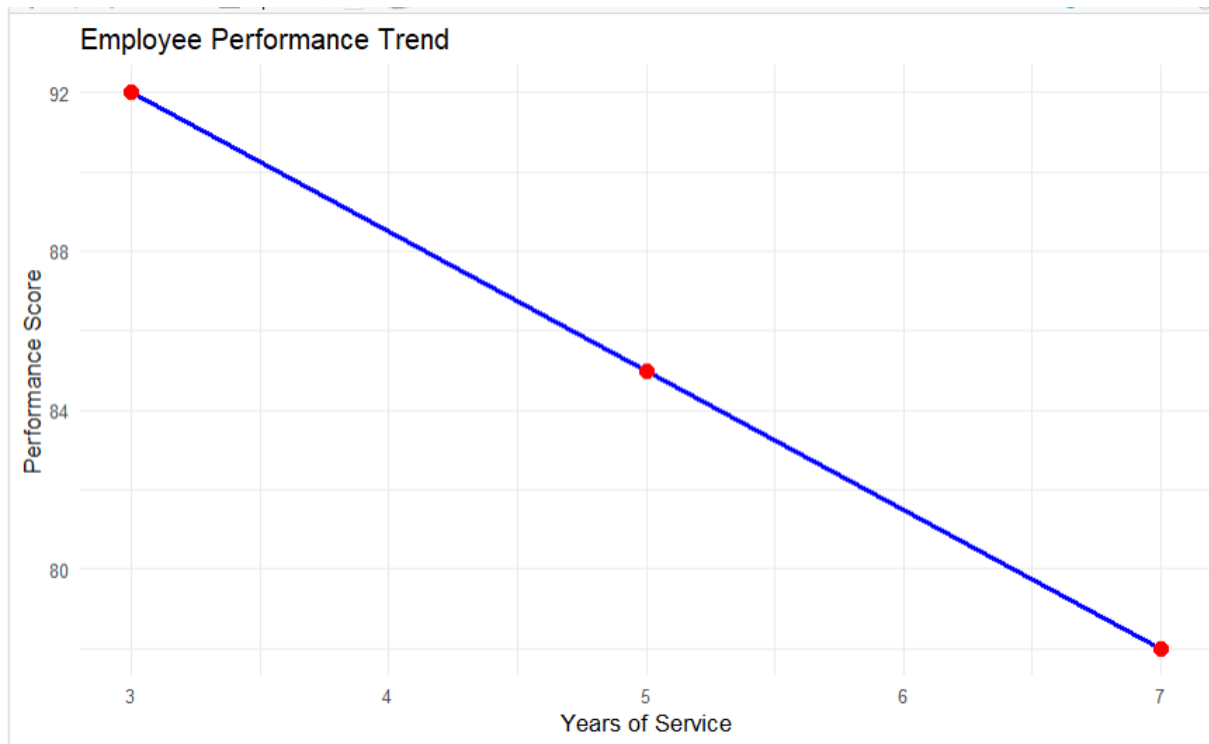
3 Marketing 7 78

### 1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
ggplot(employee,  
  aes(x = Years_of_Service,  
      y = Performance_Score,  
      group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score"
```

```
) +
```

```
theme_minimal()
```



**2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.**

```
ggplot(employee,
```

```
  aes(x = Department,
```

```
    fill = Department)) +
```

```
geom_bar() +
```

```
labs(
```

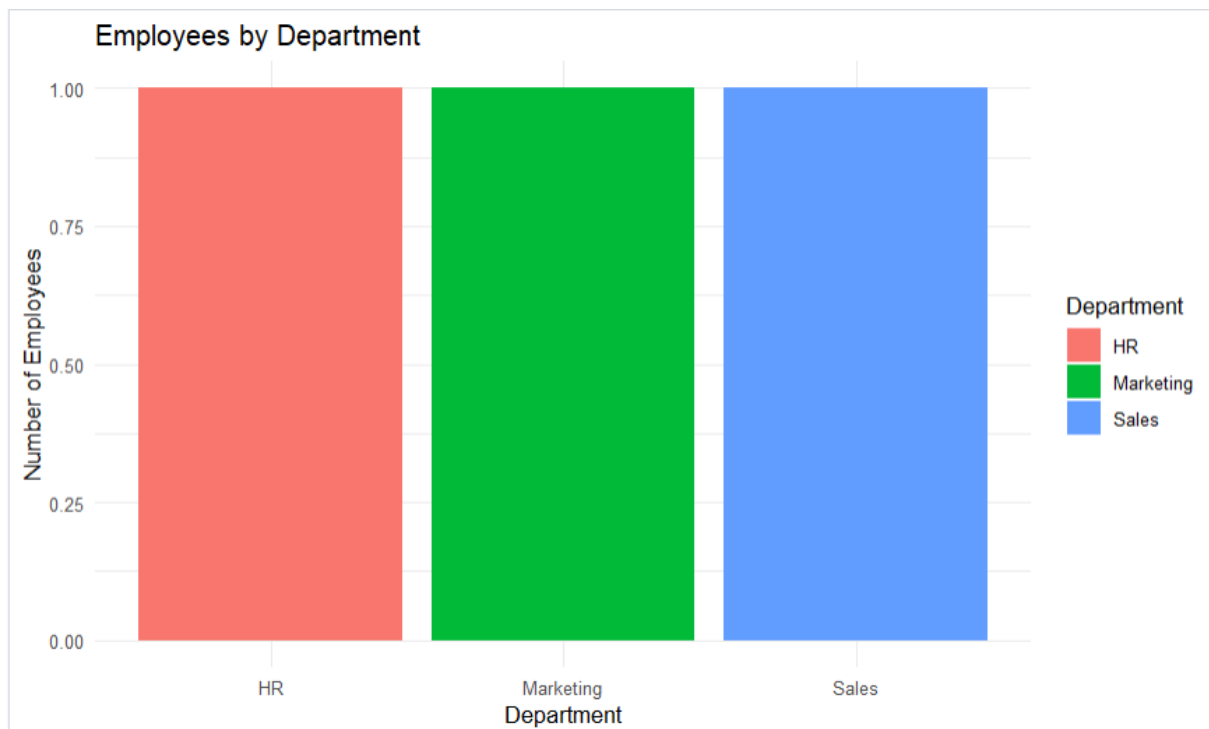
```
  title = "Employees by Department",
```

```
  x = "Department",
```

```
  y = "Number of Employees"
```

```
) +
```

```
theme_minimal()
```



**3. Build a table to display the performance data for each employee. Label the table elements.**

View(employee)

# OR

print(employee)

|   | Employee_ID | Department | Years_of_Service | Performance_Score |
|---|-------------|------------|------------------|-------------------|
| 1 | 1           | Sales      | 5                | 85                |
| 2 | 2           | HR         | 3                | 92                |
| 3 | 3           | Marketing  | 7                | 78                |

| 4 columns         |       |  |       |
|-------------------|-------|--|-------|
| Employee_ID       | <dbl> |  |       |
| [1, 3]            |       |  | 0% NA |
| Department        | <chr> |  |       |
| 3 unique          |       |  | 0% NA |
| Years_of_Service  | <dbl> |  |       |
| [3, 7]            |       |  | 0% NA |
| Performance_Score | <dbl> |  |       |
| [78, 92]          |       |  | 0% NA |